

A PROJECT TITLE

ON

Accident Hotspot Detection Using DBSCAN-Based Anomaly Detection

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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Accident Hotspot Detection Using DBSCAN-Based Anomaly Detection

1. Abstract:

Road accidents remain a serious concern across urban and rural areas, resulting in loss of life, injuries, and significant property damage, often occurring repeatedly at specific locations due to poor road conditions, congestion, sharp turns, inadequate lighting, or traffic-rule violations. Identifying such accident-prone locations manually is difficult, time-consuming, and prone to oversight. This project presents a Machine Learning-based Accident Hotspot Detection System that applies DBSCAN (Density-Based Spatial Clustering of Applications with Noise), an unsupervised clustering algorithm, to automatically discover accident hotspots from historical accident records while treating isolated incidents as anomalies. Unlike traditional clustering techniques, DBSCAN does not require the number of clusters to be specified in advance, can detect irregularly shaped clusters, and handles noisy geospatial data effectively, making it well suited for location-based accident analysis. The dataset is drawn from publicly available sources such as the US Accidents Dataset, UK Road Safety Dataset, or Indian Road Accident Dataset, and includes features such as accident latitude and longitude, date, time, road type, speed limit, weather conditions, visibility, road surface condition, junction type, traffic signal presence, accident severity, and number of vehicles involved. The workflow involves data collection, preprocessing, exploratory data analysis, feature selection of key spatial and contextual attributes, and application of the DBSCAN algorithm using tuned `eps` and `min_samples` parameters to group nearby accident locations into clusters. Clustering quality is assessed using the Silhouette Score, Davies-Bouldin Index, visual inspection, and domain expert validation. The resulting system classifies locations as high-risk hotspots, medium-risk areas, low-risk areas, or isolated anomalies, and visualizes results on interactive maps using tools such as Folium and Plotly. This project supports applications including smart city traffic management, road safety analysis, accident prevention planning, emergency response optimization, and insurance risk assessment. Future enhancements include integrating real-time accident reporting, combining live GPS, weather, and traffic data for dynamic hotspot detection, developing an interactive web or mobile application, applying deep learning for severity prediction, and recommending safer alternative routes based on detected hotspots.

2. INTRODUCTION:

Road traffic accidents continue to pose a major public safety challenge, often clustering around specific locations due to factors such as poor road design, heavy congestion, sharp turns, inadequate lighting, or frequent traffic-rule violations. Traditional approaches to identifying these accident-prone locations rely heavily on manual analysis of accident reports, which is slow, inconsistent, and difficult to scale across large regions. With the growing availability of geo-tagged accident records from government portals and open datasets, unsupervised machine learning techniques can be applied to automatically detect spatial patterns in accident occurrence. This project uses DBSCAN, a density-based clustering algorithm well suited to spatial data, to group accident locations into meaningful hotspot clusters while identifying isolated incidents as anomalies. Unlike centroid-based clustering methods, DBSCAN does not require the number of clusters to be predefined and can detect irregularly shaped, dense regions, making it particularly effective for real-world accident data. The resulting hotspot map is intended to support traffic authorities, urban planners, and emergency services in prioritizing road safety interventions and allocating resources more effectively.

3. SOFTWARE REQUIREMENTS:

Programming Language: Python 3.x

Libraries: Pandas, NumPy, Scikit-learn (DBSCAN), Matplotlib, Seaborn, Folium, Plotly

Development Environment: Jupyter Notebook / Google Colab / Visual Studio Code

Operating System: Windows / Linux / macOS

Signature of the Guide

Course Instructor